



Addis Ababa University
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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY!

NUTRACEUTICALS

Submitted to: Mr Aynalem Redae

DEFINITION

- Products derived from foods providing health benefits beyond basic nutrition.
- Contain concentrated bioactive compounds.
- Often formulated as capsules, tablets, or extracts.
- No universally accepted regulatory definition.

CATEGORIES OF NUTRACEUTICALS

- **Nutraceuticals** Classified into **functional foods**, **dietary supplements**, **medical foods**

Functional Foods

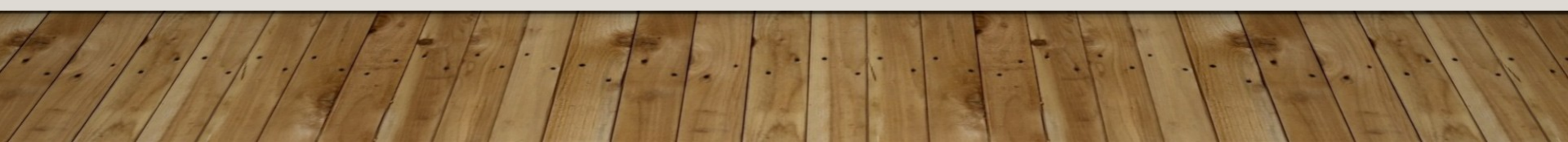
- Whole or fortified foods consumed for health benefits
- Example: Salmon, nuts → **Omega-3 fatty acids**
- Benefits: Supports brain function, cardiovascular health, reduces inflammation

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Dietary Supplements

- Pills, capsules, liquids, or powders
- Contain concentrated nutrients from foods
- Example: **Omega-3 supplements** used with or without food sources

Medical Foods

- Used under **physician supervision**
 - Help manage diseases or medical conditions
 - Example: Special foods for digestive issues, liquid nutrition for those unable to eat enough solid food
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HISTORICAL BACKGROUND: EARLY CONCEPTS

- Ancient civilizations used food for medicinal purposes.
- Hippocrates emphasized 'Let food be thy medicine.'
- Ayurvedic and Chinese medicine included dietary therapies.

HISTORICAL BACKGROUND: SCIENTIFIC ERA

- 20th century discoveries of vitamins and essential nutrients.
- Isolation of amino acids, fatty acids, phytochemicals.
- Established scientific basis for food health relationships.

LATE 20TH CENTURY: 'NUTRACEUTICAL'

- Term introduced in the late 1980s.
- Used to describe food-derived products with therapeutic benefits.
- Expansion of research on antioxidants, probiotics, and omega-3s.

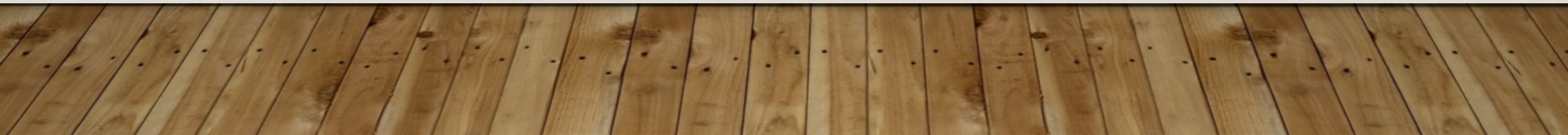
1990S–2000S: INDUSTRY GROWTH

- Rise of functional food science.
- Advances in extraction and encapsulation technologies.
- Increased consumer demand for natural health products.

CONTEMPORARY PERIOD

- Emphasis on evidence based evaluation.
- Regulatory differences across countries.
- Focus on quality control, purity, and safety.

CURRENT GLOBAL AND ETHIOPIAN PRACTICE



NUTRACEUTICAL TRENDS

- Nutraceuticals are changing the landscape of health and wellness. These products, which blur the line between food and medicine, offer a proactive approach to health management that resonates with an increasingly health-conscious global population.
- Nowadays, nutraceuticals have received considerable interest due to potential nutritional, safety and therapeutic effects. A market research recently proposed that the worldwide nutraceuticals market is expanding and would reach US \$250 billion by 2018.

FACTORS CONTRIBUTE TO THE EXPANDING NUTRACEUTICAL

Several factors contribute to the expanding nutraceutical market:

- Aging population seeking preventative health solutions
- Increased awareness of the link between diet and health
- Rising incidence of lifestyle-related diseases
- Growing interest in personalized nutrition
- Shift towards natural and organic products

POPULAR NUTRACEUTICAL PRODUCTS

- The nutraceutical category encompasses a wide range of products designed to provide health benefits beyond basic nutrition. Companies in this space must stay attuned to evolving consumer preferences and emerging product categories.
- Some of the most popular nutraceutical products include:
 - Dietary supplements (vitamins, minerals, herbal extracts)
 - Functional foods (fortified cereals, omega-3 enriched eggs)
 - Protein bars and powder

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- Probiotic-rich foods and supplements
- Energy drinks and functional beverages
- Antioxidant-rich products
- Plant-based and vegan nutritional products

Each of these categories presents unique formulation and packaging challenges, requiring innovative solutions to ensure product stability, efficacy, and consumer appeal.

PRACTICE OF NUTRACEUTICALS IN ETHIOPIA

The practice of nutraceuticals in Ethiopia is deeply rooted in traditional medicine and the use of indigenous wild edible plants (WEPs), with a growing modern regulatory framework for commercial products.

An estimated 80% of the population relies on traditional plant-based remedies for primary healthcare, highlighting their significant role in the national health system.

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- **Widespread Indigenous Knowledge:** Local communities, particularly in rural and semi-arid areas, possess extensive indigenous knowledge about plants that serve both nutritional and medicinal purposes (nutraceuticals).
- **Primary Food and Medicine Source:** WEPs are a crucial source of vitamins, minerals, protein, and antioxidants, used not just as supplemental or emergency food during shortages but also for daily health maintenance and treating various ailments like hypertension, infections, and digestive issues.

THE GLOBAL NUTRACEUTICAL MARKET

- Was estimated at USD 451.8 billion in 2024.
- The market is expected to grow from USD 480.4 billion in 2025 to USD 746.6 billion in 2034, at a CAGR of 5%,
 - ✓ North America: Largest market (tech & research leadership)
 - ✓ Europe: Fastest-growing (strong health & wellness focus)

KEY GROWTH DRIVERS

- ❖ Rising health & wellness awareness
- ❖ Preventive healthcare adoption (chronic disease focus)
- ❖ Demand for plant-based, organic, clean-label products
- ❖ Expanding middle-class in emerging markets
- ❖ Advances in personalized nutrition & functional ingredients

ETHIOPIA'S STRENGTHS

- Biodiversity: Rich in medicinal plants, herbs, and natural resources.
- Traditional knowledge: Long history of herbal medicine and dietary practices.
- Agricultural base: Potential to scale production of herbal teas, adaptogenic herbs, fortified cereals.
- Emerging middle class: Growing awareness of preventive healthcare

OPPORTUNITIES

- Export potential for herbal supplements, teas, functional foods.
- Development of local nutraceutical industries (digestive health, immunity boosters).
- Investment in value-added processing (functional beverages, fortified foods).
- Collaboration with universities (e.g., Jimma University research on food science & technology)

CHALLENGES

- Limited regulatory frameworks for nutraceuticals.
- Need for clinical validation & quality standards.
- Infrastructure gaps in processing & distribution.
- Competition from established global players.

STRATEGIC PATH FORWARD

- Strengthen policy & regulation for nutraceutical certification.
- Build research & innovation hubs (linking universities & industry).
- Promote sustainable cultivation of medicinal plants.
- Encourage public-private partnerships for investment & export.

REFERENCE

- Jimma University Journals (EJAST – agriculture, food science, herbal research)
- UNCTAD report on nutraceutical opportunities for LDCs (Ethiopia)
- <https://www.colorcon.com/education-insights/nutraceutical-trends-an-overview>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC10717384/>

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Thank You